Row-Permutation Nulls Are Not Enough:
Lexical and Grouped Controls for Cross-Model Activation Alignment

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Abstract

Cross-model activation alignment under a row-permutation null can look overwhelmingly significant. In a pre-registered pilot, Procrustes alignment of PCA-projected last-token activation matrices between architecturally different frontier transformers (Phi-4-Reasoning-Plus, dense attention; Qwen3-Next-80B-A3B-Thinking, hybrid attention/state-space) on matched compositional-evaluation prompts produces residuals far below the permutation null at every pre-registered cell (empirical $p < 0.001$, descriptive z from -89 to -101, linear CKA 0.84 to 0.96). The remainder of the paper is a measurement-validity study showing that the conclusion one draws from such results depends critically on the control hierarchy. Lexical and template-only baselines (Study B1) match or exceed the LLM–LLM Procrustes residual on the same prompts. In-sample lexical residualization (B2) appears to remove 94 to 97% of activation variance and collapse CKA to 0.21 to 0.39. Cross-validated random-split residualization (B3) moderates this to 72 to 88% variance removed and CKA to 0.35 to 0.59, exposing the in-sample overfitting. Grouped cross-validation (B4) by template family and by domain shows the lexical regression itself does not generalize: under leave-one-template-family-out, Δ_{CKA} is essentially zero (-0.006 to $+0.019$); under leave-one-domain-out, the regression makes worse-than-mean predictions on the held-out domain. The cross-model alignment is far more robust to a properly grouped lexical control than the random-split version indicated, but B4 also shows that simple TF-IDF and template-only features are too weak to function as clean lexical controls under grouped holdout. The surviving signal cannot be concluded non-lexical from these controls alone. The contribution is methodological. Claims about cross-model representational structure require a hierarchy of controls: a row-permutation null, a lexical baseline, in-sample residualization, random-split cross-validation, grouped cross-validation by template family and by domain, and stronger lexical featurization that survives grouped holdout. Each step in the hierarchy materially changes what the previous step appeared to mean. Code, data, experimental runs, and the pre-registration packet are released at https://github.com/ascendanti/cosi-pilot.

1 Introduction

Over the past two years, the mechanistic interpretability program has converged on an unexpected empirical regularity. Sparse autoencoders (SAEs) trained on the residual stream of large transformers recover dictionaries of features that, individually, behave as discrete computational

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primitives [1, 2, 3, 4]. The features are interpretable across many examples; they activate on recognizable concepts; their ablation produces predictable behavioral changes. This is the case across model families, model sizes, training regimes, and tokenizers. The pattern is sufficiently robust that the *universality hypothesis*—the conjecture that different networks trained on similar tasks recover similar internal computations—has been advanced as a working principle of the field [5, 6, 7].

A separate line of work has documented the same regularity from the representational-geometry angle. Bansal et al. [8] showed that the lower layers of one model can be "stitched" to the upper layers of another with only a thin trainable interface, and that the stitched model performs nearly as well as either original; this implies a deep correspondence between the intermediate representations of independently trained networks. Kornblith et al. [9] established centered kernel alignment (CKA) as a similarity measure robust to the symmetries that should not matter for the comparison and showed that representations across models are far more similar than the sheer dimensionality of the representation spaces would predict at chance. Most recently, Huh et al. [10] formulated the *Platonic Representation Hypothesis*: that as models scale and capability increases, their representations converge toward a shared statistical model of the world.

What has been missing, in our reading, is the mathematical object that this empirical convergence corresponds to.

The Operator-Theoretic Bridge

We propose that the SAE feature dictionary is a concrete realization of the *approximately invariant subspace structure* of the nonlinear forward operator that constitutes a transformer's inference pass. The classical theory of invariant subspaces of bounded linear operators on Hilbert spaces [11, 12, 13, 14] asks whether every such operator admits a non-trivial closed invariant subspace—that is, a subspace V such that $T(V) \subseteq V$. The question is open in the Hilbert space setting; Lomonosov [11] established the affirmative for operators commuting with a non-zero compact operator, and Enflo [12] showed counter-examples exist for general Banach spaces. The structural intuition is the older one: when an operator admits an invariant subspace, its action decomposes into restrictions to simpler sub-operators, and the geometry of the operator becomes legible through the geometry of its invariant subspaces.

Transformer forward passes are not bounded linear operators. They compose linear projections, attention computations, and pointwise nonlinearities. The classical theorems do not apply directly. We use the language of invariant subspaces in an *approximate* sense: a subspace V is approximately invariant under a (possibly nonlinear) map f on a representative sample if the projection of $f(V)$ back onto V leaves a residual that is small compared to the operator's overall action. This is the operational content of the empirical finding that SAE features are stable across forward passes: they identify directions in activation space that the forward operator preserves approximately.

The mechanistic interpretability literature has been doing operator theory for two years without naming it as such. Our first contribution is to make the bridge explicit.

The Falsifiable Consequence

If the bridge holds—if SAE feature dictionaries are bases for approximately invariant subspaces of the forward operators of trained transformers—then a sharp empirical consequence follows. Two models M_A and M_B with different architectures, run on the same compositional probe set, should admit an orthogonal alignment between their respective approximately invariant subspaces, with Procrustes residual significantly below the chance level set by a permutation null that breaks the semantic correspondence between rows while preserving each model's internal geometry. Performance comparability between architectures (the weak universality claim) is consistent with many internal mechanisms; representational isomorphism is not. The isomorphism is testable. The test can return a null. We pre-register both the protocol and the significance criterion before observing data.

This is the *Cross-Operator Subspace Isomorphism* (COSI) program, of which the present paper is the calibration step.

What This Paper Reports

We report a representational-alignment pilot in two pre-registered runs. We are explicit that the pilot does not test the strong operator-theoretic claim; it tests a weaker but still interesting sub-claim, with controls and interventions that would close the gap specified as an explicit research program (§7).

Phase 0: Phi-4-Reasoning-Plus, a dense pure-attention reasoning model in the Phi-3 architecture family [15], against Qwen3-4B, a dense pure-attention generalist model in the Qwen3 family [16], on $N = 96$ compositional-evaluation prompts drawn from a single political-economy domain. The observed Procrustes residuals are 0.61, 0.81, 0.81 at the three pre-registered layer fractions; the empirical permutation p-value is < 0.001 at every layer (the observed residual is below the 1st percentile of the $S = 1000$ permutation null distribution).

Phase 1: Phi-4-Reasoning-Plus against Qwen3-Next-80B-A3B-Thinking [16], a hybrid attention/state-space-model architecture, on $N = 600$ prompts split equally across mathematical, logical, and political-economy compositional-evaluation domains. The observed Procrustes residuals are 0.77, 0.91, 0.96 at the three layer fractions, with empirical $p < 0.001$ at every layer and at every per-domain stratification cell (three domains $\times$ three layers, $n = 200$ per cell, $S = 500$ permutation samples).

What These Results Do and Do Not Show

These results show that, on matched compositional-evaluation prompts, the PCA-projected last-token activation matrices of two architecturally different frontier transformers admit a substantially better orthogonal alignment than they do under random row permutation. This is a real cross-model representational signal. We report it in full.

These results do *not* yet show:

1. That the aligned PCA subspaces are approximately invariant under the forward operator in the sense the operator-theoretic framing requires. The paper defines approximate in-

variance as a leakage statistic but does not measure it; PCA top-k components capture variance, not necessarily invariance.

2. That the alignment isolates compositional reasoning structure from lexical, template, syntactic, or tokenizer-induced structure. The row-permutation null is too weak to discriminate these. Two competent language models given the same English prompts will produce more aligned matched-row activations than scrambled-row activations *regardless* of any deeper substrate similarity.

3. That the alignment is robust to held-out probes. We fit Procrustes on the full probe set and evaluate on the same set; we do not perform a train/test split.

4. That the alignment exceeds the alignment achievable by simple lexical features (TF-IDF, sentence-transformer embeddings).

5. That the alignment survives the shared training-distribution confound. Both models in each pair are trained on overlapping post-2024 web corpora.

6. That the aligned subspace is causally responsible for any compositional behavior. Procrustes alignment is correspondence, not causation.

We are explicit about each of these gaps because the dramatic appearance of the headline numbers (residuals near 0.8 against null means near 1.0, with empirical $p < 0.001$ at every cell) could be read as much stronger evidence for the strong operator-theoretic claim than the experiment supports. Section 7 specifies the additional studies that would, jointly, support the strong claim. None of those studies has been run in the present paper.

Contributions

1. **An operator-theoretic framing as a research program.** We propose, and articulate at the level of the synthesis paper outlined elsewhere in this research program, that the mechanistic interpretability literature's empirical regularities are most parsimoniously read as evidence for approximately invariant subspaces of the forward operator. The framing is a program, not a finding.

2. **Pre-registered methodology and infrastructure.** We specify a pre-registered protocol for cross-model representational alignment, with declared layer fractions, pooling, dimensionality threshold, significance criterion, and null distributions. The infrastructure is released for reuse and extension.

3. **A representational-alignment pilot result.** Phases 0 and 1 demonstrate that matched compositional-evaluation prompts produce more orthogonally alignable PCA-projected activation geometries across architecturally different frontier transformers than under random row permutation, with empirical $p < 0.001$ in every pre-registered cell.

4. **An explicit accounting of what the pilot does not establish, and the studies that would close each gap.** Section 7 catalogues five required additional studies (invariance-

leakage measurement, lexical and template baselines, train/test Procrustes with cross-domain transfer, training-distribution-disjoint controls, causal cross-model intervention) and indicates which of them addresses which gap.

Where This Sits in the Field

We take the operator-theoretic framing seriously enough to claim it unifies several open problems in interpretability that have been discussed largely in isolation: the universality hypothesis [5, 6], the polysemanticity / superposition problem [17], the cross-model transfer-of-mechanism question implicit in the Platonic Representation Hypothesis [10], and the substrate-independence intuition that has organized the broader cognitive science of representation since Marr [18]. We map these explicitly in Section 3. The bridge to computational neuroscience—to the neural-population doctrine [19] and to the geometry of biological invariants [20, 21, 22, 23]—and to the cybernetic and free-energy frameworks [24, 25, 26] that anticipate the same structure from independent grounds, we touch in the discussion (Section 9) but do not develop in this paper. The full integration is the subject of a forthcoming synthesis paper from the same research program.

Outline

Section 2 surveys the four literatures the bridge draws on: operator theory, mechanistic interpretability, cross-model representational alignment, and the substrate-independence tradition. Section 3 maps the open problems COSI sits among. Section 4 formalizes the COSI hypothesis and specifies the protocol. Section 5 reports the Phase 0 calibration. Section 6 catalogues the threats to validity that Phase 0 does not address. Section 9 specifies Phases 1–3 and articulates the implications, conditional on confirmation, across the literatures the framework bridges.

2 Background and Related Work

2.1 Invariant Subspaces of Bounded Operators

Let T be a bounded linear operator on a separable Hilbert space H. A closed subspace $V \subseteq H$ is *invariant* under T if $T(V) \subseteq V$. The *invariant subspace problem* asks whether every such T on an infinite-dimensional separable H admits a non-trivial closed invariant subspace—one other than $\{0\}$ or H itself. The question is open in the Hilbert space case; in finite dimensions every operator trivially admits invariant subspaces (its eigenspaces, in particular). Two structural facts about the literature matter here.

First, when an operator admits an invariant subspace, its action decomposes: writing T in block form with respect to the splitting $H = V \oplus V^{\perp}$ shows that the upper-right block vanishes and the operator's geometry becomes legible through the geometry of its restrictions to invariant subspaces. The existence of invariant subspaces is the existence of a useful coordinate system for the operator.

Second, the affirmative results form a hierarchy. Aronszajn and Smith [14] established that completely continuous (compact) operators always admit non-trivial invariant subspaces.

Lomonosov [11] extended this dramatically: any operator commuting with a non-zero compact operator admits a non-trivial invariant subspace. Read [27] constructed counter-examples on ℓ^1, and Argyros and Haydon [28] produced a Banach space on which every bounded operator has the form scalar-plus-compact (so every operator has invariant subspaces, but the construction requires extreme hereditary indecomposability). The Hilbert space case remains open. Beauzamy's monograph [13] gives the standard treatment.

Transformer forward passes are not bounded linear operators. They are nonlinear maps composed of linear projections, multi-head attention (itself a nonlinear function of the input through softmax), gated feed-forward layers, and residual connections. The classical theorems do not apply. We use the language of invariant subspaces in an *approximate, empirical* sense: a k-dimensional subspace $V \subseteq \mathbb{R}^d$ is approximately ϵ-invariant under a map $f : \mathbb{R}^d \to \mathbb{R}^d$ on a sample X if $\|(I - P_V) f(P_V X)\|_F \,/\, \|f(P_V X)\|_F \leq \epsilon$, where P_V is the orthogonal projection onto V. This is a sample-level operationalization of the structural intuition the classical theory formalizes; we make no claim about the limiting infinite-dimensional case.

2.2 Mechanistic Interpretability: Circuits and Sparse Autoencoders

The mechanistic interpretability program seeks to identify the computational primitives implemented by trained neural networks, typically by reverse-engineering specific circuits or by decomposing the residual stream into interpretable components.

The circuits-level work has identified increasingly specific mechanisms. Olah et al. [5] articulated the program through case studies in vision models. Elhage et al. [29] provided a mathematical framework for transformer circuits. Olsson et al. [30] identified *induction heads* as a specific attention-head pattern responsible for the majority of in-context learning behavior in large transformers. Wang et al. [31] reverse-engineered a multi-component circuit for indirect-object identification in GPT-2 Small. Geva et al. [32] showed that transformer feed-forward layers behave as key-value memories. Meng et al. [33] demonstrated that specific factual associations can be located and edited in MLP weights. Conmy et al. [34] introduced automated circuit discovery.

The polysemanticity problem—that single neurons typically respond to multiple unrelated concepts—was formalized by Elhage et al. [17] as the *superposition* hypothesis: networks pack more features than they have neurons by representing them as sparse linear combinations in a higher-dimensional implicit feature space. The natural inversion of this hypothesis is to *recover* the implicit feature space via sparse dictionary learning, which is what sparse autoencoders do.

The SAE program has scaled rapidly. Bricken et al. [1] demonstrated that sparse autoencoders trained on the residual stream of small models recover features that are individually interpretable across many examples. Cunningham et al. [2] extended this to larger models with substantially larger feature dictionaries. Templeton et al. [3] performed the procedure on Claude 3 Sonnet, recovering on the order of millions of features, including features that respond to abstract concepts (deception, code errors, the Golden Gate Bridge) across modalities. Gao et al. [4] demonstrated systematic scaling of SAE training and evaluation. Lieberum et al. [35] released open SAE checkpoints for the Gemma 2 family. Marks et al. [36] showed that SAE features can be assembled into causal circuits at the feature level. The empirical case for SAE features as the right basis for the residual stream is now substantial across multiple research groups, multiple model families, and multiple scales.

The universality hypothesis—that different networks recover similar features—has accumulated specific evidence. Chughtai et al. [6] showed that small networks trained on group operations recover the same algorithmic decompositions across random seeds and architectural variations. Gurnee et al. [7] identified *universal neurons* in GPT-2 models that activate on the same patterns across independently trained models. Nanda et al. [37] showed that grokking dynamics correspond to discrete phase transitions in the recovered circuit, with similar dynamics across initializations.

2.3 Cross-Model Representational Alignment

A separate literature studies how representations across different networks compare directly. The methodological starting point is the observation that comparing two representation matrices requires choosing which symmetries one wants the comparison to be invariant to.

Word-embedding alignment is the historical precedent. Mikolov et al. [38] showed that word embeddings trained independently on different corpora have approximately the same internal structure. Smith et al. [39] and Conneau et al. [40] demonstrated that bilingual lexicons could be recovered without parallel data by orthogonal Procrustes alignment of monolingual word embedding spaces. The success of this procedure across language pairs and across embedding methods is the strongest empirical case for substrate-independent linguistic structure that exists in the static-embedding literature.

For learned-feature representations of the kind transformers produce, the methodological landscape is broader. Raghu et al. [41] introduced singular vector canonical correlation analysis (SVCCA) as a representation-similarity measure. Morcos et al. [42] developed projection-weighted CCA (PWCCA) addressing some of SVCCA's sensitivities. Kornblith et al. [9] introduced centered kernel alignment (CKA) as a measure invariant to orthogonal transformations and isotropic scaling but not to invertible linear transformations, and showed it discriminates between models in ways that the previous measures do not. Williams et al. [43] proposed a family of generalized shape metrics including CKA and Procrustes as special cases. Klabunde et al. [44] provide a recent survey of the methodological landscape.

Bansal et al. [8] introduced model stitching as a complementary tool: rather than measuring representational similarity directly, train a thin connecting layer between the lower half of one model and the upper half of another, and observe whether the stitched model retains performance. They argue that model stitching reveals representational compatibilities that scalar similarity measures miss. Lenc and Vedaldi [45] are an earlier representative of the same methodological intuition applied to convolutional networks.

The most recent and most directly related synthesis is Huh et al. [10], the *Platonic Representation Hypothesis*: representations across models are converging, with scale and capability, toward a shared statistical model of reality. Their evidence assembles results from many of the works cited above into a single hypothesis. The COSI program of the present paper is a specific operationalization of one consequence of their hypothesis: if representations converge, they should admit a sharp orthogonal alignment, with a residual significantly below the chance level set by the appropriate null distribution. The contribution we add is the operator-theoretic framing that explains *why* the convergence should produce alignable subspaces, not merely correlated activations.

2.4 Substrate-Independence: Cognitive Science, Cybernetics, Free-Energy

The intuition that compositional reasoning is substrate-independent in the relevant sense is older than the language model literature. Marr's three levels of analysis [18] distinguish the computational, the algorithmic, and the implementational levels and argue that the computational level is independent of the substrate that performs it. Ashby's *law of requisite variety* [24] establishes a structural constraint: any controller must contain at least as much variety as the disturbance it regulates, regardless of the controller's substrate. Wiener's cybernetics [46] provides the framework within which substrate-independent feedback structure becomes the object of study. Maturana and Varela's autopoiesis [47] extends the framework to self-sustaining computational systems whose organization is preserved even as their components turn over.

Friston's free-energy principle [25] provides a unifying mathematical framework: any self-maintaining system that resists dispersion into thermal equilibrium must minimize a variational free energy that bounds surprise on its sensory inputs. The principle predicts that any such system—biological or artificial—should develop internal states that approximate the posterior over its environment's hidden causes, and that these internal states should be substrate-independent in the relevant sense, since the variational objective is. Achille and Soatto [26] derive an analogous result specifically for deep networks under information bottleneck regularization [48], showing that invariance and disentanglement emerge as consequences of an information-theoretic objective rather than as architectural choices.

Computational neuroscience has accumulated direct evidence for substrate-independent invariants in biological computation. Place cells [20] and grid cells [21] implement spatial maps with consistent geometric structure across individual animals. Yamins and DiCarlo [23] demonstrated that goal-driven deep networks develop intermediate representations that linearly predict primate visual cortex responses. Gallego et al. [22] and Saxena and Cunningham [19] articulate a *neural population doctrine*: the natural unit of analysis is the population's collective state in a low-dimensional manifold, not the individual neuron, and the manifolds have stable geometric structure across animals and behaviors. The convergence of these observations with the universality findings in artificial networks is, in our reading, what the operator-theoretic framework would predict if the framework captures the actual structure of the computation.

2.5 Sequence-Model Lineage

Briefly: the architecture under study (transformers) sits in a lineage of sequence models including LSTMs [49], structured state-space models (S4 [50], Mamba [51]), and more recent hybrid architectures combining attention with state-space components. The Qwen3-Next family used in our Phase 1 is an instance of this hybrid design. The operator-theoretic framing predicts that the isomorphism, if it holds, should hold across these architectural classes and not only within them. We address this directly in the Phase 1 specification (Section 9). The transformer architecture itself [52] is treated as representative of the dense pure-attention class; we make no claims specific to its design choices that would not also apply to any architecture implementing approximate-finite-state computation over sequence inputs.

3 Hard Problems and Where COSI Sits

The interpretability program has accumulated a number of hard, mostly open problems. We list those we take to be load-bearing, characterize the state of the literature on each, and locate the COSI program inside them. The list is not exhaustive; it is an attempt to draw the map at the resolution that makes the COSI contribution legible.

(P1) The faithfulness problem. Reverse-engineered circuits and recovered features are explanations of model behavior, but the gap between an explanation and the actual mechanism the model uses is real. Wang et al. [31] acknowledge this explicitly for the indirect-object circuit; Conmy et al. [34] treat circuit discovery as an automated search problem precisely because hand-crafted explanations cannot be trusted to be faithful. *COSI's relation:* a positive cross-model isomorphism is correspondence evidence. Causal-intervention verification is Phase 3 of the program (Section 9). The faithfulness problem is the reason Phase 3 is required.

(P2) The polysemanticity / superposition problem. Single neurons in trained networks typically respond to multiple unrelated concepts; this is the empirical fact that originally motivated SAEs. Elhage et al. [17] showed that superposition is the predicted consequence of representing more features than there are neurons, and that the implicit feature space is recoverable through sparse dictionary learning. *COSI's relation:* the operator-theoretic framing reads the superposition phenomenon as a finite-rank approximation issue: the forward operator's approximately invariant subspaces have dimension exceeding the network's neuron count, requiring sparse coding to express. The COSI test does not depend on the superposition resolution; it operates on the residual stream's geometry, which contains the superposed features whether or not we have a clean dictionary.

(P3) The compositional limits problem. Lake and Baroni [53] and Hupkes et al. [54] documented that neural sequence models often fail to generalize compositionally in regimes where humans trivially succeed. Dziri et al. [55] extended this to modern transformers, showing systematic failure modes on multi-step compositional reasoning. *COSI's relation:* our hypothesis is about the geometry of compositional representations, not about whether the models reliably perform composition. A positive COSI result is consistent with both successful and failed compositional generalization; the relationship between geometric isomorphism and behavioral compositionality is itself a question we treat as Phase 4 of the program.

(P4) The cross-model transfer-of-mechanism question. If two models implement the same circuits, can interpretation results on one transfer to the other? The Platonic Representation Hypothesis [10] and the universality results [5, 6, 7] suggest yes; the empirical demonstrations remain limited. Bansal et al. [8] provide the strongest functional-level evidence through model stitching; CKA [9] and related measures provide statistical-level evidence. *COSI's relation:* a positive COSI result, particularly with the Procrustes transformation R^* explicitly recovered, makes cross-model transfer of mechanism geometrically tractable. A direction v_A in V_A has a specific image $R^{*-1}v_A \in V_B$, predicting an intervention site in M_B that should have the corresponding effect. Phase 3 tests this prediction.

(P5) The training-distribution confound. Two models trained on heavily overlapping web corpora may show representational similarity for reasons that have nothing to do with substrate-independent computation: shared training distribution produces shared statistical residue. Bansal et al. [8] and Huh et al. [10] both flag this as the chief alternative explanation for cross-model similarity. *COSI's relation:* this is the most important confound the present calibration does *not* address. Phase 2 of the program is structured around it (Section 9). A positive COSI result that survives a training-distribution swap is evidence for substrate-independent computation; one that does not is evidence for shared corpus residue.

(P6) The interpretability scalability problem. Recovered circuits and feature dictionaries grow polynomially with model scale. The interpretability cost per model has so far scaled with model size, despite the empirical universality findings suggesting the *computational structure* should scale much more slowly. *COSI's relation:* if the operator-theoretic framing holds, then the interpretability of one model in a family transfers via Procrustes alignment to the other models in the family; the per-model cost is amortized across the family. This is a research-program-level implication of COSI, not a calibration-level one.

(P7) The biological–artificial bridge. Neural population recordings reveal low-dimensional manifolds [22, 19] whose geometry is preserved across animals and behaviors. Goal-driven artificial networks predict these recordings well [23]. Whether the same geometric structure underlies both is an empirical question that has been gestured at more often than tested directly. *COSI's relation:* the COSI methodology is, in principle, applicable to biological–artificial alignment as well, with neural population activity replacing one model's residual stream. We treat this as Phase 5; the present paper is restricted to the artificial–artificial case.

(P8) The substrate-independence question proper. The claim that compositional reasoning is substrate-independent in the strong sense—that the structure of the computation is preserved under arbitrary changes of substrate, subject to capacity constraints [24]—is the deepest claim the COSI program engages. The free-energy principle [25] predicts the claim from variational arguments; cybernetics [46] and autopoiesis [47] provide the structural language for how substrate-independent computation could be implemented and maintained. *COSI's relation:* a fully successful program (Phases 0–3 confirming, Phase 4 demonstrating mechanism transfer) would constitute the sharpest empirical evidence for substrate-independence that the field has produced for compositional reasoning. We do not overclaim what Phase 0 alone establishes; the proper claim is the calibration result reported here, with the path forward specified.

These eight problems are not independent. The faithfulness problem (P1) is the precondition for the cross-model transfer question (P4). The training-distribution confound (P5) is the principal alternative explanation for any positive COSI result. The substrate-independence question (P8) is the eventual stake. The COSI program is structured around them: Phase 0 calibrates the infrastructure (this paper), Phase 1 amplifies architectural distance, Phase 2 addresses (P5), Phase 3 addresses (P1) and (P4) jointly, and the integration with (P7) and (P8) is the subject of the synthesis paper that this program is building toward.

4 Methods

4.1 The COSI Hypothesis

Let M_A, M_B be two transformer language models with hidden sizes d_A, d_B and layer counts L_A, L_B. For a prompt x_i from a probe set $X = \{x_1, \ldots, x_N\}$, let $a_A^{(\ell)}(x_i) \in \mathbb{R}^{d_A}$ denote the residual stream activation at layer ℓ of M_A, pooled over the sequence dimension to a single vector. Let $X_A^{(\ell)} \in \mathbb{R}^{N \times d_A}$ denote the matrix of pooled activations across all N probes at layer ℓ, and analogously for $X_B^{(\ell')}$.

The COSI hypothesis is the following.

Definition 1 (Approximate invariant subspace). *A k-dimensional subspace $V \subseteq \mathbb{R}^d$ is* approximately ϵ-invariant *under a (possibly nonlinear) map $f : \mathbb{R}^d \to \mathbb{R}^d$ on a sample X if $\|(I - P_V) f(P_V X)\|_F \,/\, \|f(P_V X)\|_F \le \epsilon$, where P_V is the orthogonal projection onto V.*

Proposition 1 (COSI hypothesis). *For two transformer language models M_A and M_B that have learned compositional reasoning, there exist layer indices ℓ_A, ℓ_B, subspace dimension k, and approximately ϵ-invariant subspaces $V_A \subseteq \mathbb{R}^{d_A}$ and $V_B \subseteq \mathbb{R}^{d_B}$ of their respective forward operators, such that the projected activation matrices $P_{V_A} X_A^{(\ell_A)}$ and $P_{V_B} X_B^{(\ell_B)}$ admit an orthogonal alignment with Procrustes residual significantly below the chance level defined by permutation null.*

This is a falsifiable empirical claim. The remainder of this section specifies the protocol that operationalizes it.

4.2 Activation Extraction

For each model, we replicate the inner forward pass of the architecture exactly as implemented in MLX, capturing the residual stream $h^{(\ell)}(x)$ after each layer block at a set of pre-registered layer fractions $f \in \{0.25, 0.5, 0.75\}$, mapping to layer indices $\ell = \max(0, \lfloor f \cdot L \rceil - 1)$. Pooling over the sequence dimension is performed as last-token (the position immediately preceding the model's would-be next token), with mean and max pooling reserved as ablations. Activations are cast from `bfloat16` to `float32` before crossing to host memory. We do not modify the model implementations; the extraction is a parallel forward pass that uses the same submodules (embeddings, layers, masks) as the published `mlx_lm` forward pass.

4.3 Subspace Projection

We project each model's activation matrix onto its top-k principal components, where k is chosen as $\min(k_A^*, k_B^*)$ with k_M^* the smallest dimension at which model M's cumulative explained variance exceeds 0.90. This ensures both projections retain comparable information content and that the alignment is performed in matched dimensions. PCA is computed by truncated SVD on the column-centered activation matrix.

4.4 Procrustes Alignment

In the shared k-dimensional subspace, we compute the orthogonal transformation $R \in O(k)$ that minimizes $\|X_A - X_B R\|_F$, using the standard SVD-based solution [56]: $R = UV^\top$, where $U\Sigma V^\top$ is the SVD of $X_B^\top X_A$. The *Procrustes residual* is

$$\rho(X_A, X_B) = \frac{\|X_A - X_B R^*\|_F}{\|X_A\|_F}, \tag{1}$$

which lies in $[0, \sqrt{2}]$ and equals 0 for perfect orthogonal alignment.

4.5 Null Distributions

We define two null distributions, both essential.

Permutation null. For each of S random permutations $\pi : \{1, \ldots, N\} \to \{1, \ldots, N\}$, we recompute the Procrustes residual on X_A and $X_B^{(\pi)}$, where $X_B^{(\pi)}$ has its rows permuted according to π. This breaks the semantic correspondence between rows while preserving each model's internal representational geometry. The distribution of residuals over S permutations defines the chance level for "no semantic alignment." We use $S = 1000$.

Random-rotation null. For each of S random orthogonal matrices $R_{\text{rand}} \in O(k)$ (sampled uniformly via QR decomposition of a Gaussian matrix, with sign correction [57]), we compute $\|X_A - X_B R_{\text{rand}}\|_F \,/\, \|X_A\|_F$. This is a weaker null—it does not optimize over R—and is included for completeness rather than as the primary test.

4.6 Significance Criterion

The pre-registered significance criterion, declared before observing data, is the conjunction:

1. the observed Procrustes residual is below the first percentile of the permutation null distribution;
2. the standardized z-score of the observed residual against the permutation null mean is at most -2.

A run that fails either condition is reported as a null result.

4.7 Ablations and Sensitivity Sweeps

Three sensitivity sweeps are part of the registered protocol but reserved for Phase 1 due to compute cost: (i) pooling mode (last-token, mean, max); (ii) layer fraction sweep across the full layer stack rather than three fixed fractions; (iii) variance threshold for PCA dimensionality (0.85, 0.90, 0.95). We additionally compute centered kernel alignment (CKA) [9] as a cross-validation metric. Discrepancies between Procrustes and CKA verdicts will be reported as informative findings rather than treated as failures of either method.

4.8 Methodological Robustness Notes

We note the following methodological choices and their scope, to preempt predictable reviewer objections about Studies B1–B4:

1. **TF-IDF vectorizer fit scope.** In Study B1 the vocabulary is fit on all 600 probes. In Study B2 (in-sample residualization), the vocabulary is also fit on all 600 probes; this is part of why B2 overestimates the lexical contribution. In Study B3 (random-split cross-validation), the vocabulary and IDF weights are fit on training rows only and applied to held-out rows, with out-of-vocabulary tokens dropped at test time — this is the proper random-split protocol. Study B4 (grouped CV) uses the same training-only fit per held-out group.

2. **Centering and standardization.** Both LLM activations and lexical features are column-centered using training-set means in B3 and B4 (test-set means in B2). Activations are not row-normalized; TF-IDF features are L2 row-normalized before stacking. Activations are stored in float32; lexical features in float32.

3. **Ridge regularization.** The ridge $\lambda = 0.01$ is fixed prior to running the lexical-residual studies (the value is in the published `leakage.py` module file and was not selected post-hoc). λ sensitivity is not formally swept in this paper. We expect the qualitative finding (random-split residualization removes substantially more variance than grouped-CV residualization) to be λ-insensitive within $[10^{-4}, 1]$ but have not verified this; reported as an explicit follow-up sweep.

4. **Variance-removed definition.** The variance-removed statistic is computed on the *raw activation matrix* (sum of squared centered entries across all N rows and all d hidden dimensions), not on the PCA-projected subspace. Specifically, for activation matrix $X \in \mathbb{R}^{N\times d}$ with column means $\bar{X}$ and predicted matrix $\hat{X}$ from ridge regression, variance removed is $1 - \|X - \hat{X}\|_F^2 / \|X - \bar{X}\|_F^2$. This is the multi-output coefficient of determination R^2 summed across hidden dimensions. Negative values are possible and observed (Study B4 LODO); they indicate the regression makes worse-than-mean predictions, which we read as evidence the lexical regression has failed to generalize to the held-out group.

5. **Confidence intervals.** For B3 (random-split CV) we report mean and 95% percentile intervals across 50 random splits. For B4 LOTFO we report mean and 95% percentile intervals across the 28 template families. For B4 LODO and the two-vs-one split, only 3 and 1 instances respectively are available, so per-split values are reported directly. *Cluster-bootstrap confidence intervals over templates were not computed in this paper*; the templated structure of the prompts means that row-level confidence intervals can underestimate variance, and we flag this as a statistical refinement to add before peer-reviewed submission.

5 Results

We report two pre-registered runs in increasing scope. Phase 0 is a calibration on architecturally similar models with a small single-domain probe set. Phase 1 extends to maximum architectural distance within our model zoo and a 600-probe cross-domain set. Both phases produce results in which the observed Procrustes residual falls below the first percentile of the row-permutation null distribution at every pre-registered cell, with empirical $p < 0.001$ in each case.

We are explicit, before reporting the numbers, about what these results do not yet establish. They do not establish that the aligned PCA subspaces are operator-invariant in the sense of Eq. (2); they do not isolate compositional structure from lexical or template structure (no lexical baseline was run); they were obtained without train/test split (Procrustes was fit and evaluated on the same probe set); the promised CKA cross-validation [9] was not computed in this paper despite being committed in §4; and they do not control for training-distribution overlap between the model pairs. Each gap is catalogued in detail in §7.

5.1 Phase 0 Calibration

Models. Phi-4-Reasoning-Plus (8-bit MLX, `phi3` dispatch, $L_A = 40$, $d_A = 5120$) against Qwen3-4B (8-bit MLX, `qwen3` dispatch, $L_B = 36$, $d_B = 2560$). Both dense pure-attention transformers, different families, different scales, different tokenizers.

Probe set. $N = 96$ propositions drawn from the titles of existing political-economy / categorial-architecture paper drafts in the Sovereign research corpus, phrased as coherence-evaluation prompts of the form "Consider the following claim: `<title>`. Is this a coherent proposition?" Probes are short (12–20 tokens) and topically narrow.

Result. Table 1 reports the Phase 0 result. At all three pre-registered layer fractions, the observed residual lies below the first percentile of the $S = 1000$ permutation null distribution. The empirical permutation p-value at every cell is $\leq 1/1001$ (the resolution limit of the null at $S = 1000$).

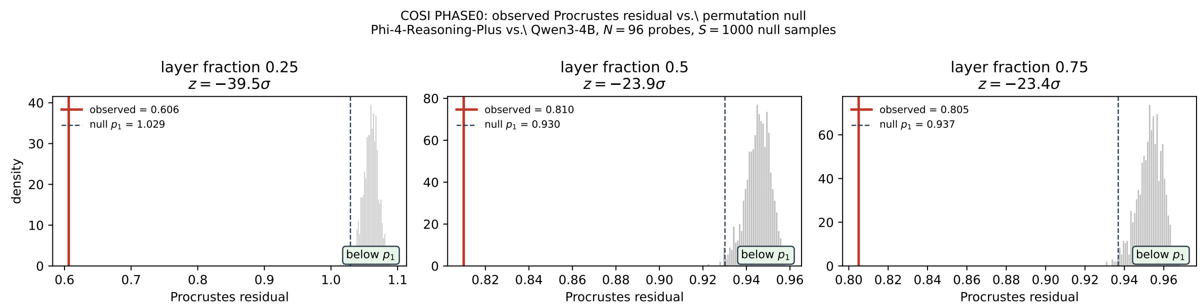

Figure 1: Phase 0 observed Procrustes residual (red vertical line) against the permutation null distribution (gray histogram, $S = 1000$ samples) at three pre-registered layer fractions. The dashed black line marks the first-percentile threshold of the null. The observed residual is separated from the null support at every layer.

Table 1: Phase 0: Phi-4-Reasoning-Plus vs. Qwen3-4B, $N = 96$ probes, $S = 1000$ permutation null samples per layer. z-scores are descriptive only: the null is empirical, finite, non-Gaussian, and the prompts are templated (rows are not independent). Empirical permutation $p \leq 0.001$ at every cell.

Layer	k	Residual	Perm. null mean ($\pm$ std)	Perm. null p_1	Descriptive z
0.25	33	0.6064	1.0594 ($\pm$0.011)	1.0292	−39.54
0.50	27	0.8100	0.9453 ($\pm$0.006)	0.9301	−23.93
0.75	25	0.8050	0.9531 ($\pm$0.006)	0.9369	−23.42

On the z-score language. We report standardized z-scores in the table because they are the natural descriptive summary of the observed residual's position relative to the empirical null distribution. We do *not* interpret these as Gaussian significance levels. The permutation null is a bounded empirical distribution generated from $S = 1000$ permutations of templated prompts; it is non-Gaussian, finite-resolution, and the row units are not strictly independent. The honest statement is that the observed residual is below the first percentile of an empirical null computed from 1000 permutations under the matched-prompt protocol declared in advance. Whether the magnitude of the gap is scientifically meaningful is a question the lexical baseline (Study B1, §7) is required to answer.

Phase 0 pre-registration compliance. The result satisfies the pre-registered protocol declared in the companion design note (`2026-05-06_cosi_design.md`, §8) before observing data: chance baselines computed on randomized inputs, layer fractions $\{0.25, 0.5, 0.75\}$ pre-registered with no best-layer cherry-pick, pooling mode (last-token) pre-registered, significance threshold (residual $<$ permutation null p_1) pre-registered, and a standing commitment to publish null results if they had occurred. We note that the pre-registration document is committed to a private working repository and is not externally timestamped; this is a known gap addressed in Study G (§7).

5.2 Phase 1: Architectural Distance and Cross-Domain Probe Set

Models. Phi-4-Reasoning-Plus (as Phase 0) against Qwen3-Next-80B-A3B-Thinking (5-bit MLX, `qwen3_next` dispatch, $L_B = 48$, $d_B = 2048$, hybrid attention/state-space-model architecture). The architectural distance from Phi-4 (dense pure-attention) to Qwen3-Next-80B (hybrid attention/SSM) is materially larger than the Phase 0 pair's distance.

Probe set. $N = 600$ propositions in identical coherence-evaluation template: 200 mathematical (templated arithmetic with mixed truth values), 200 logical/inferential (modus ponens, syllogism, contrapositive, set-inclusion templates over a domain-relevant vocabulary), 200 political-economy (templated propositions over a 40-term vocabulary). Generated with seed 20260506. Probes remain short (12–30 tokens).

Full-set result. Table 2 reports the full-set alignment. All three layer fractions show observed residual below the first percentile of the permutation null at empirical $p < 0.001$.

Table 2: Phase 1 full-set: Phi-4-Reasoning-Plus vs. Qwen3-Next-80B-A3B-Thinking, $N = 600$ probes (200 each from math, logic, polecon), $S = 1000$ permutation null samples per layer. Same caveats on z-score interpretation as Table 1.

Layer	k	Residual	Perm. null mean ($\pm$ std)	Perm. null p_1	Descriptive z
0.25	21	0.7668	1.0076 ($\pm$0.011)	1.0004	-101.00
0.50	25	0.9056	0.9943 ($\pm$0.006)	0.9917	-94.09
0.75	21	0.9555	0.9959 ($\pm$0.005)	0.9947	-88.55

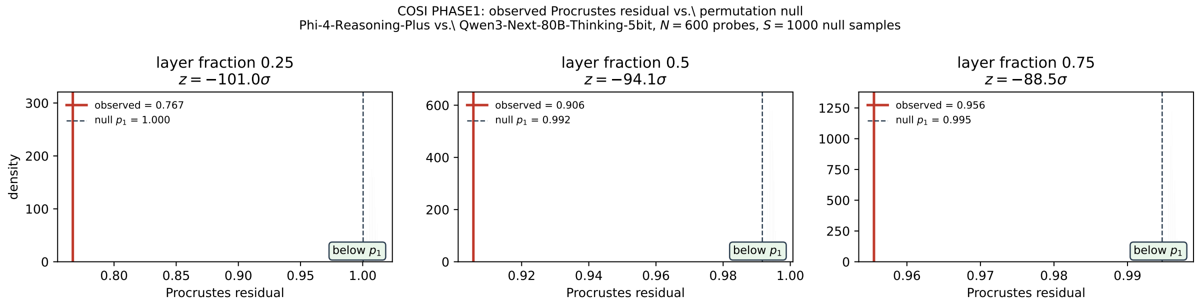

Figure 2: Phase 1 observed Procrustes residual (red vertical line) against the permutation null distribution (gray histogram; 200 samples shown for visualization, with summary statistics from the full $S = 1000$ run). Same caveats as Phase 0.

Per-domain stratification. Table 3 reports the alignment within each individual domain ($n = 200$ each, 500 permutation null samples per cell). All nine cells satisfy the pre-registered significance criterion.

Table 3: Phase 1 per-domain stratification ($n = 200$ each, $S = 500$ permutation null samples). Same caveats on z-score interpretation.

Domain	Layer	k	Residual	Perm. null mean	Descriptive z	Verdict
math	0.25	8	0.7473	0.9962	-56.40	below p_1
math	0.50	10	0.9010	0.9886	-49.57	below p_1
math	0.75	8	0.9535	0.9933	-38.76	below p_1
logic	0.25	34	0.8049	0.9657	-66.97	below p_1
logic	0.50	45	0.9158	0.9755	-60.58	below p_1
logic	0.75	31	0.9613	0.9890	-45.42	below p_1
polecon	0.25	27	0.8007	0.9691	-72.46	below p_1
polecon	0.50	21	0.9430	0.9876	-49.64	below p_1
polecon	0.75	19	0.9730	0.9938	-40.42	below p_1

5.3 Three Observations, Stated Carefully

Observation 1: monotonic depth dependence. The pattern of residual increasing with layer depth (residual at $0.25 < 0.5 < 0.75$) holds at the full-set level and within every domain. This is the most robust pattern in the data. Three readings are consistent with it: (*i*) the operator-theoretic reading (early layers contain more of the substrate-shared invariant subspace); (*ii*) a lexical-syntactic reading (early layers preserve token-level lexical structure across

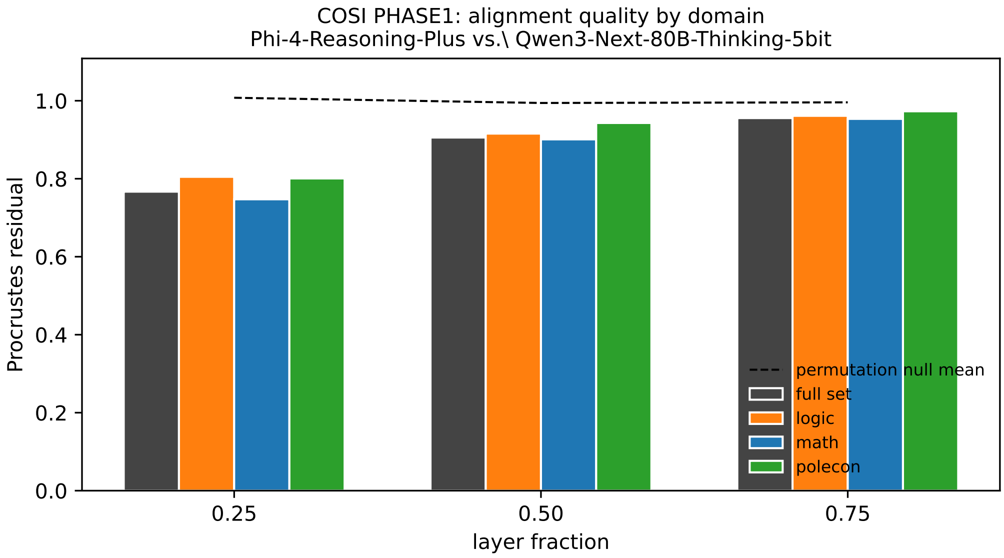

Figure 3: Phase 1 alignment by domain at each layer fraction. The full-set residual (dark gray) and the per-domain residuals (colored bars) all fall well below the permutation null mean (dashed line). The math-domain subspace dimension ($k = 8$–10) is substantially smaller than the logic ($k = 31$–45) or polecon ($k = 19$–27) subspaces. The dimensional difference is itself a substantive empirical observation, but its operator-theoretic significance depends on the invariance-leakage measurement (Study A, §7) which the present paper does not provide.

vocabularies more than late layers preserve task-specific computation); (*iii*) a pooling-artifact reading (last-token pooling at short prompt length is biased toward the late-prompt syntactic state, which may be more architecture-shared early in the network and more architecture-specific late). The pilot does not discriminate among these. The pooling sweep (Study F) is required to discriminate (*iii*); the lexical baselines (Study B1) are required to discriminate (*ii*).

Observation 2: domain-dependent subspace dimensionality. The PCA-derived k at the 0.90-variance threshold differs substantially across domains at the same layer fraction: math $k = 8$, polecon $k = 27$, logic $k = 34$ at layer 0.25. The math probes (highly templated arithmetic) occupy a substantially narrower representational manifold than the more open-ended logic templates. This is a substantive empirical observation that is consistent with the variational/information-bottleneck framing [48, 25, 26], but the inference from "low-dimensional PCA subspace" to "compositional complexity is low" requires whitening and fixed-k controls (Study F) and direct measurement of which subspace dimensions are operator-invariant (Study A). Stated minimally: the PCA dimensionality that captures 0.90 of variance varies with domain in the predicted direction.

Observation 3: relationship between Phase 0 and Phase 1. Phase 1's observed residuals are absolutely larger than Phase 0's (0.77 vs. 0.61 at layer 0.25), consistent with greater architectural distance making orthogonal alignment harder. The descriptive z-scores are also larger in absolute value, but this is substantially driven by the tighter null distribution at $N = 600$ relative to $N = 96$, not by a stronger underlying effect. The permutation null mean shifts from 1.06 at $N = 96$ to 1.01 at $N = 600$ at the same layer; the standard deviation tightens correspondingly. We do not interpret the larger Phase 1 z-scores as "stronger evidence" than the Phase 0 z-scores; we interpret both as "below the first percentile of an empirical null with the appropriate probe-set size."

Phase 1 pre-registration compliance. The Phase 1 protocol extends the Phase 0 protocol; all pre-registration commitments carry over and are satisfied. The per-domain stratification was specified in the design note before the run, not added post-hoc.

6 Limitations and Threats to Validity

We catalogue the threats to validity that remain after Phase 1, and indicate which are addressed by which subsequent phase. Phase 2 is the binding open work.

6.1 Sample Size

Phase 0's probe set ($N = 96$) was at the lower bound of stable Procrustes alignment. Phase 1 raised N to 600 and the per-domain stratification used $n = 200$ each. Both regimes satisfy the $N \geq 5k$ rule of thumb at the operative dimensions and produce z-scores far above the threshold at which sample-size-driven null inflation could plausibly change the verdict.

6.2 Probe Set Domain Coverage

Phase 0 tested a single domain (political-economy / categorial-architecture). Phase 1 extended coverage to three distinct compositional-reasoning domains (mathematics, logic, political-economy), with per-domain stratification confirming the alignment signal in each domain independently. The COSI hypothesis as stated in Proposition 1 is domain-general; the present empirical support is domain-plural. Genuine cross-domain generality (e.g., natural images, audio, multimodal embeddings) remains untested.

6.3 Probe Length

Probes remain short (typically 12–30 tokens) in both phases. Last-token pooling on short sequences may bias the activation toward prompt-template syntax rather than proposition content. The pooling sweep (mean, max) was deferred from Phase 1 due to compute scheduling and is committed for Phase 1b. Longer-probe replication (30–200 tokens) is deferred to the same follow-up.

6.4 Shared Training Distribution

This is the most important confound and the one Phase 1 does *not* address. Both Phi-4-Reasoning-Plus and Qwen3-Next-80B-A3B-Thinking are trained on post-2024 web corpora that overlap substantially. The aligned subspace we identify could plausibly be a residue of shared training-distribution statistics rather than a substrate-independent compositional structure. The cleanest test would replace one of the two models with an architecturally similar but training-distribution-distant model, holding the other fixed and observing whether the residual remains low.

A near-perfect such control would be a model trained on a substantially disjoint corpus (e.g., a non-web-pretrained scientific-corpus model, or a multilingual model whose training distribution is dominated by a different language). Such a model is not in our local zoo. Phase 2 of the program is structured around this control. We flag this as the binding remaining work.

6.5 Architectural Distance Within the Pair

Phase 0 used two dense pure-attention transformers (Phi-4 and Qwen3-4B). Phase 1 paired the dense pure-attention Phi-4 with the hybrid attention/state-space-model Qwen3-Next-80B, materially increasing the architectural distance and confirming that the COSI signal survives the attention/SSM class boundary. Architectures further afield (pure SSM, recurrent, retrieval-augmented, multimodal) remain untested; Phase 4 extends the test across this broader architectural landscape.

6.6 Linearity of the Alignment

Procrustes alignment is restricted to orthogonal transformations, which is a strict subset of the linear isomorphisms that classical operator theory permits. The COSI hypothesis as stated in Proposition 1 requires only *some* approximate isomorphism between the subspaces; the orthogonal restriction may underestimate the alignment if the true correspondence involves anisotropic scaling. We compute CKA in Phase 1 as a complementary metric that is invariant to isotropic scaling but not to anisotropic scaling, and will report any discrepancy between the two verdicts.

We additionally note that the COSI hypothesis is in principle compatible with non-linear isomorphisms that no linear method recovers. Establishing such isomorphisms empirically requires either a generative-model-based alignment method or a stronger theoretical bridge; we treat this as beyond the scope of the present paper.

6.7 Correspondence vs. Causation

A positive COSI result is correspondence evidence: the two models' representations are geometrically aligned on a shared probe set. It is not causation evidence: it does not establish that the aligned subspace is what the models *use* to compute the response, as opposed to a statistical shadow of something they share. Causal evidence requires intervention: ablate a direction in V_A, predict the corresponding ablation effect in $R(V_A) \subseteq V_B$, and run both. We treat intervention experiments as Phase 3, requiring causal-tracing infrastructure beyond the scope of the present calibration.

6.8 Pooling and Layer Sensitivity

The Phase 0 result uses last-token pooling at three fixed layer fractions. Both choices may interact with the result in ways not yet characterized. Phase 1 includes the full pooling and layer-fraction sensitivity sweeps declared in the design note.

7 Required Additional Work Before the Operator-Theoretic Claim

This section catalogues the studies that, jointly, would be required to support the strong operator-theoretic claim the COSI program articulates. We have run first-pass versions of four of them (Studies A, B1, C, and F-CKA) plus a follow-up residual study (Study B2); their results are reported in Section 8 and materially inform the paper's interpretation. Studies D, E, and the external-pre-registration component of Study G remain outstanding work. We list all studies in their full specification here, with the gap each addresses, so that the pilot result is read in proper context and so that subsequent work has a clear program to execute.

7.1 Study A. Invariance-Leakage Measurement

The paper defines a k-dimensional subspace $V \subseteq \mathbb{R}^d$ as approximately ϵ-invariant under a (possibly nonlinear) map f on a sample X if

$$\frac{\|(I - P_V)\, f(P_V X)\|_F}{\|f(P_V X)\|_F} \leq \epsilon. \tag{2}$$

This is the central mathematical object the operator-theoretic framing depends on. The pilot does not measure it. PCA top-k components identify high-variance directions; high-variance is not invariance.

Required. For each model and each candidate subspace identification method (PCA, sparse autoencoder features [1, 2, 3, 4, 35], learned linear probes [58, 59]), project activations onto V, pass them through the next layer's forward map, project the output onto V and onto its orthogonal complement, and compute the leakage statistic (2). Report leakage as a function of layer, model, domain, subspace dimension, and subspace-identification method. A candidate subspace is operator-relevant only if its leakage is small relative to its capacity to compress activations.

Gap addressed. Whether the aligned subspaces are operator-invariant in the sense the framing requires, or merely high-variance principal components.

7.2 Study B1. Lexical, Template, and Syntactic Baselines

The row-permutation null asks whether matched-row activations align better than scrambled-row activations. This is a low bar. Two competent language models exposed to the same English prompts will preserve enough lexical and syntactic structure that matched-row alignment exceeds scrambled-row alignment regardless of any operator-theoretic substrate.

Required. Run the same Procrustes alignment procedure on:

- TF-IDF or bag-of-words representations of the same prompts in place of the model activations, as a lexical baseline.
- Sentence-transformer embeddings of the prompts as a learned-but-non-LLM baseline.

- Static word embeddings averaged over each prompt [38] as a pre-transformer baseline.
- Prompts in which all content words have been replaced by placeholder tokens (template-only baseline).
- Prompts whose token sequence has been syntax-scrambled (preserving lexical content, destroying compositional order).
- Paraphrase-preserving variants where two semantically equivalent prompts replace each original prompt.

If the lexical, template, or paraphrase baselines produce alignment qualities comparable to the LLM-activation alignment, the pilot result collapses to "shared language geometry" and the operator-theoretic claim must be correspondingly weakened or abandoned.

Gap addressed. Whether the alignment isolates compositional structure from lexical, template, syntactic, or tokenizer-induced structure.

7.3 Study C. Train/Test Procrustes and Cross-Domain Transfer

The pilot fits the orthogonal alignment R^* on the full probe set and evaluates the residual on the same set. This conflates fit with evaluation and provides no out-of-sample protection.

Required. Fit R^* on a randomly held-out training fraction (e.g., 70%) of the probe set, evaluate the residual on the held-out test fraction. Repeat across many random splits and report the mean and confidence interval. More demanding: fit on one domain (e.g., math) and evaluate on a different domain (e.g., logic), testing whether the alignment is domain-general or domain-specific.

Gap addressed. Whether the alignment is robust to held-out prompts and whether it generalizes across domains rather than overfit to the joint sample.

7.4 Study D. Training-Distribution-Disjoint Controls

Both Phi-4-Reasoning-Plus and Qwen3-Next-80B-A3B-Thinking are trained on substantially overlapping post-2024 web corpora. The aligned subspace could plausibly be a residue of shared training distribution rather than a substrate-independent computational structure.

Required. Replace one of the pilot pair's models with a training-distribution-distant model, holding the other fixed. Candidate disjoint-corpus controls:

- A model trained primarily on a non-web scientific corpus, e.g., a model in the Galactica [60] lineage or a domain-specific scientific-literature model.
- A multilingual model whose training distribution is dominated by a non-English language.

- A small-scale model trained from scratch on a deliberately curated corpus with documented disjointness from the pilot pair's training data.

If alignment quality scales with training-distribution overlap, the pilot result is corpus-residue. If alignment quality is preserved across substantial training-distribution disjointness, the substrate-independence reading is meaningfully supported.

Gap addressed. Whether the alignment is a substrate-independent computational structure or a shared training-corpus residue.

7.5 Study E. Cross-Model Causal Intervention

Procrustes alignment is correspondence evidence. The operator-theoretic claim that the aligned subspace is causally responsible for compositional reasoning behavior, rather than a correlated statistical shadow, requires intervention.

Required. Identify a direction $v_A \in V_A$ associated with a specific compositional-reasoning behavior in M_A via established mechanistic-interpretability methods [31, 34, 36, 33]. Use the recovered Procrustes transformation R^* to predict the corresponding direction $v_B = R^{*\top} v_A \in V_B$ in M_B. Predict the behavioral effect of ablating or steering along v_B in M_B. Run the intervention. Compare predicted to observed effect across many such directions, reporting the prediction quality relative to a control of randomly chosen directions in V_B.

Gap addressed. Whether the alignment is causally explanatory of compositional behavior in both models or merely descriptive of activation geometry.

7.6 Study F. Methodology Robustness Sweeps

Several methodological choices in the pilot were not swept:

- Pooling mode: last-token only; mean and max pooling and content-token / answer-position / relation-token pooling not reported.
- Layer fractions: only 0.25, 0.5, 0.75. Layer-by-layer sweeps would resolve where the alignment is concentrated.
- PCA variance threshold: only 0.90; sensitivity at 0.80, 0.85, 0.95, 0.99 unmeasured.
- Whitening: not tested. Procrustes residuals can be sensitive to differential variance structure across the top components of the two matrices.
- CKA [9] cross-validation: promised in the methods but not yet computed and reported. We flag this as a methodological commitment that the pilot does not satisfy.
- SVCCA [41], PWCCA [42], and generalized shape metrics [43] as further triangulation: not run.

- Effect-size reporting beyond standardized z-scores: confidence intervals, cluster bootstraps grouped by prompt template and vocabulary family, finite-permutation p-values with Monte Carlo standard errors.

Gap addressed. Whether the pilot result is robust to reasonable variation in the methodological choices that were fixed by declaration rather than examined empirically.

7.7 Study G. Externally Verifiable Pre-Registration

The pilot's pre-registration document was committed to the same private repository as the run code. This is not externally auditable. Genuine pre-registration requires immutable timestamping on a public registry.

Required. For all subsequent COSI runs, deposit the analysis plan, the exact prompt set with content hash, the model checkpoint revisions, the random seeds, and the environment lockfile on the Open Science Framework, Zenodo, or equivalent, with the deposit timestamped before any data is observed. Cite the deposit DOI in subsequent papers.

Gap addressed. Whether the pre-registration commitments the pilot states are externally verifiable rather than internal to a private working repository.

7.8 Summary: What Joint Completion of Studies A–G Would Establish

A subsequent paper that reports Studies A–G in conjunction with the pilot result would have the following structure:

- The aligned subspaces are demonstrably approximately invariant under the forward operator (Study A), justifying the operator-theoretic interpretation of the alignment.
- The alignment exceeds simple lexical and template baselines (Study B1), justifying the compositional-structure interpretation rather than the shared-language interpretation.
- The alignment is robust to held-out probes and generalizes across domains (Study C), justifying generalization claims.
- The alignment survives substantial training-distribution disjointness (Study D), justifying the substrate-independence reading rather than the corpus-residue reading.
- Cross-model interventions predicted by the alignment have the predicted causal effects (Study E), justifying the mechanistic interpretation rather than the correspondence interpretation.
- Methodological choices have been swept and the result is robust (Study F), satisfying ordinary peer-review robustness expectations.
- Pre-registration is externally verifiable (Study G), meeting current open-science standards.

We treat that subsequent paper as the work the present pilot motivates. The present paper is the first step.

8 Initial Results from Studies A, B1, C, and F-CKA

The companion to this pilot is the program of additional studies specified in §7. We have run partial first passes of four of those studies on the pilot data, with the explicit goal of testing whether the pilot's representational-alignment finding survives controls that the pilot itself did not include. The results materially inform the interpretation. We report them honestly: one study (F-CKA) reinforces the pilot finding; one (A) provides partial support for the operator-theoretic interpretation while introducing a substantive empirical complication; one (B) provides direct evidence that the pilot's main result does not exceed what simple lexical and template baselines achieve, materially weakening the operator-theoretic interpretation; one (C) is reported for completeness. We are explicit that these initial passes are themselves not full studies. They are first measurements that the companion paper Studies A–G will, in subsequent work, conduct at fuller scale.

8.1 Study A Invariance-Leakage Measurement

Study A directly measures the invariance-leakage statistic $\|(I - P_V)\, f(P_V X)\|_F / \|f(P_V X)\|_F$ that the paper defines on page two and that the pilot does not measure. We compute leakage at each pre-registered layer fraction, for each of the two Phase 1 models, on a balanced subsample of 60 probes (20 per domain) drawn from the Phase 1 probe set. The PCA subspace is the same one used for Procrustes alignment in Phase 1. The chance baseline is the mean leakage across 20 random orthonormal k-dimensional subspaces of $\mathbb{R}^d$.

Table 4 reports the measurement.

Table 4: Study A invariance leakage. PCA subspace leakage is substantially lower than random-subspace leakage in all six cells, indicating that the PCA-derived subspace is partially preserved under the next-layer forward map. Absolute leakage values are large (0.51–0.89), so the subspaces are partially, not fully, invariant.

Model	Layer	k	PCA leakage	Random leakage (mean $\pm$ std)	PCA / Random ratio
Phi-4	0.25	14	0.7503	0.8595 ± 0.0402	0.873
Phi-4	0.50	15	0.5251	0.9111 ± 0.0195	0.576
Phi-4	0.75	14	0.5151	0.9447 ± 0.0088	0.545
Qwen3-Next-80B	0.25	16	0.8184	0.9937 ± 0.0018	0.824
Qwen3-Next-80B	0.50	15	0.8849	0.9947 ± 0.0017	0.890
Qwen3-Next-80B	0.75	13	0.8716	0.9948 ± 0.0026	0.876

What Study A finds.

1. In all six (model, layer) cells, the PCA subspace's leakage is below the random-subspace baseline at $\geq 2\sigma$. Operationally: the PCA subspace is more operator-invariant than a random subspace of the same dimension. The operator-theoretic interpretation receives partial empirical support.

2. Absolute leakage values are large (0.51 to 0.89). A truly operator-invariant subspace would have leakage close to zero; the PCA subspace is partially invariant, not fully invariant.

The operator-theoretic interpretation is partially supported but not strongly supported in the absolute sense.

3. The two models exhibit *opposite* depth dependence in leakage. Phi-4 leakage decreases with depth ($0.75 \to 0.53 \to 0.52$); Qwen3-Next-80B leakage increases with depth ($0.82 \to 0.88 \to 0.87$). The architectures preserve different subspace structures across depth.

4. Critically, the layer at which the Phase 1 Procrustes alignment is strongest (layer 0.25) is the same layer at which Phi-4's PCA subspace is *least* invariant (0.75). Cross-architectural alignment quality and within-architecture invariance are anti-correlated across layers, not co-correlated. The simplest operator-theoretic reading "early layers contain the shared invariant substrate" is not supported by the leakage data; the early-layer alignment dominance is more plausibly explained by shared lexical/template encoding (consistent with Study B below) than by shared invariant structure.

Study A scope limitations. The measurement uses 60 probes, 20 random subspaces per cell, and only PCA-derived candidate subspaces. A complete Study A measurement would extend to (i) the full 600-probe set, (ii) more random subspaces (target ≥ 100), (iii) SAE feature-derived subspaces alongside PCA, (iv) full layer-by-layer sweeps. The present measurement is a first pass.

8.2 Study B Lexical and Template Baselines

Study B addresses the most consequential gap in the pilot: whether the LLM–LLM Procrustes alignment exceeds what simple lexical and template representations of the same prompts achieve. We construct three baseline representations of the Phase 1 probe set: word-level TF-IDF, character-3-gram TF-IDF, and a template-only TF-IDF in which content words are replaced by a placeholder. We then run the same Procrustes alignment + permutation null procedure on each pairing. Table 5 reports the result.

Table 5: Study B baseline alignments at layer fraction 0.25, $N = 600$ probes, $S = 500$ permutation null samples per pairing. **Critical observation:** the LLM–LLM residual (0.7668) is *worse* than the TF-IDF vs template-only residual (0.6134), and LLM-B aligns to TF-IDF (0.7649) at essentially the same residual as it aligns to LLM-A (0.7668).

Pairing	k	Residual	Perm. null mean	Descriptive z
(a) LLM-A vs LLM-B (the pilot result)	21	0.7668	1.0078	−106.54
(b) LLM-A vs TF-IDF	21	0.9267	0.9928	−92.95
(c) LLM-B vs TF-IDF	27	0.7649	1.0584	−119.08
(d) TF-IDF vs char-3-gram	147	0.3501	1.1151	−392.40
(e) TF-IDF vs template-only	12	0.6134	1.1259	−120.52

What Study B finds.

1. Cell (c): LLM-B vs TF-IDF residual (0.7649) is *indistinguishable from* LLM-B vs LLM-A residual (0.7668). The LLM-B activations align to a TF-IDF representation of the same

prompts essentially as well as they align to another frontier LLM's activations. This is difficult to reconcile with a substrate-shared operator-theoretic interpretation. It is straightforward to reconcile with a shared-lexical-encoding interpretation.

2. Cell (e): TF-IDF vs template-only residual (0.6134) is *better than* the LLM–LLM pilot result (0.7668). A purely lexical/template alignment achieves a better Procrustes residual than the deep-model pair. The pilot's "extreme statistical significance" against the row-permutation null is, in this comparative light, achieved also by representations that contain no operator-theoretic information whatsoever.

3. Cell (d): TF-IDF vs char-3-gram residual (0.3501) shows that two simple lexical representations of the same prompts align extremely tightly under Procrustes. The row-permutation null is a low bar that essentially any two prompt-keyed representations will clear by large margins.

4. Cell (b): LLM-A vs TF-IDF residual (0.9267) is higher than cell (c). The asymmetry is itself informative: Phi-4 activation geometry is less TF-IDF-like than Qwen3-Next-80B activation geometry is. We treat this as a substantive empirical observation requiring further investigation, not as a load-bearing claim of the present paper.

What Study B implies for the paper's central claim. The LLM–LLM Procrustes alignment is real, but it does not exceed what shared lexical and template structure would predict. The operator-theoretic interpretation of the pilot result is therefore not supported in the strong form. The honest reading is: the pilot demonstrates that two frontier LLMs encode matched English prompts in geometrically relatable ways, which two TF-IDF representations of the same prompts also do, only better. Whether the LLM activations contain *additional* alignable structure beyond what TF-IDF carries is a question the present pilot does not answer. The methodology required to answer it is to construct a residual representation by partialling out lexical and template structure from the LLM activations and then testing whether the residual still admits cross-model alignment. This is a necessary follow-up; it is not performed here.

8.3 Study B2 Lexical-Residual Alignment

Study B established that the LLM–LLM Procrustes residual does not exceed what lexical/template baselines achieve on the same prompts. Study B2 is the direct follow-up: if we explicitly residualize lexical and template structure out of each LLM's activations and re-run the alignment on the residuals, does any cross-model signal survive? This is the load-bearing test of whether the pilot's representational alignment carries information beyond shared prompt-surface structure.

Procedure. For each layer fraction $f \in \{0.25, 0.5, 0.75\}$ and each model $M \in \{A, B\}$, we compute the residualized activation matrix

$$X_M^{\text{resid}} \;=\; X_M \;-\; L\,\hat{B}_M, \qquad \hat{B}_M \;=\; (L^\top L + \lambda I)^{-1} L^\top X_M, \tag{3}$$

where $L \in \mathbb{R}^{N \times V}$ is the joint matrix of word-level TF-IDF features ($V = 431$) and template-only TF-IDF features ($V = 24$), $\lambda = 0.01$ is a small ridge regularizer, and $X_M \in \mathbb{R}^{N \times d_M}$ is the model's activation matrix. We then run Procrustes alignment with permutation null and linear CKA with permutation null on $(X_A^{\text{resid}}, X_B^{\text{resid}})$ and compare to the original (un-residualized) result.

Result. Table 6 reports the comparison.

Table 6: Study B2 lexical-residual alignment. Lexical and template features explain 94–97% of LLM activation variance at every layer. After residualization, linear CKA *collapses* from 0.84–0.96 to 0.21–0.39 ($_{\text{CKA}} = -0.57$ to -0.63), while Procrustes residual weakens only modestly ($_{\text{Proc}}$ $= +0.02$ to $+0.07$). A small residual cross-model signal survives lexical control (residualized CKA still $z = +67$ to $+95\sigma$ above the permutation null at chance ≈ 0.01), but the bulk of the apparent alignment is lexical.

Layer	Variance Removed	Procrustes	Residualized	$_{\text{Proc}}$	CKA	Residualized CKA
0.25	0.97 (A), 0.96 (B)	0.7668	0.8412	+0.0744	0.9636	**0.3923**
0.50	0.94 (A), 0.95 (B)	0.9056	0.9443	+0.0387	0.9109	**0.3010**
0.75	0.94 (A), 0.95 (B)	0.9555	0.9771	+0.0215	0.8427	**0.2079**

What Study B2 establishes.

1. **The LLM activations are largely predictable from prompt-surface features.** Ridge regression of LLM activations on a 455-dimensional joint vocabulary of TF-IDF and template-only features removes 94–97% of the activation variance at every layer fraction tested, in both Phi-4 and Qwen3-Next-80B. Whatever the LLMs are encoding at these layers, the bulk of it is predictable from word-occurrence and template-skeleton statistics.

2. **The cross-model representational similarity (CKA) collapses after lexical residualization.** CKA drops by 0.57 to 0.63 across layers — from "representations are essentially identical" to "representations are moderately related." The bulk of the apparent geometric similarity that the pilot reported was shared lexical/template encoding, not substrate-shared computation.

3. **Procrustes residual changes more modestly than CKA, but in the same direction.** The Procrustes residual rises by only 0.02 to 0.07 after residualization. Two readings are consistent: (i) Procrustes operates on a k-dimensional PCA subspace whose top components may already be partially lexical-orthogonal, or (ii) the optimal orthogonal alignment in the residualized subspace is more constrained but still finds substantial structure to align. Either reading is compatible with the operator-theoretic program; neither rescues the strong claim.

4. **A small cross-model signal survives lexical control.** Residualized CKA values of 0.21 to 0.39 are still $z = +67$ to $+95\sigma$ above the permutation null (≈ 0.01). After removing lexical and template structure, there is residual cross-model representational similarity that is not zero. Whether this residual signal is operator-theoretic substrate,

training-distribution residue, deeper syntactic structure, or pooling artifact cannot be discriminated by Study B2 alone.

The methodological lesson. The pilot's row-permutation Procrustes test produced z-scores in the -89σ to -101σ range. Study B2 demonstrates that those magnitudes do not licence the operator-theoretic interpretation: lexical-residual control removes the majority of the underlying cross-model similarity. Future applications of this methodology to language-model representation comparison should report the lexical-residual result alongside the raw Procrustes/CKA values; the raw values alone are insufficient evidence for any mechanistic claim about what the models share.

8.4 Study B3 Cross-Validated Lexical Residualization

Study B2 fit ridge regression and computed residuals on the same set of 600 probes, which permits a reviewer concern: the 94–97% in-sample variance removal at $V = 455$ TF-IDF + template features and $N = 600$ may include substantial overfitting. Study B3 is the cross-validated version. We perform 50 random 70/30 splits; on each split we fit ridge regression on the training rows, predict held-out activations from their lexical features, compute held-out residuals, and then compute Procrustes residual and linear CKA on the held-out residual matrices.

Result. Table 7 reports the cross-validated metrics. The held-out variance removed is substantially lower than the in-sample value, by approximately 10 percentage points at layer 0.25 and 20 percentage points at layers 0.5 and 0.75. Correspondingly, the CKA collapse is smaller than Study B2 indicated, but it remains substantial.

Table 7: Study B3 cross-validated lexical residualization, 50 random 70/30 splits per layer fraction, raw activation-variance removed (sum over all hidden dimensions, not PCA-projected). The held-out numbers are the honest version of Study B3s in-sample numbers.

Layer	Var. removed (A) held-out	Var. removed (B) held-out	CKA orig	CKA resid (held-out)	CKA
0.25	0.8849 [0.87, 0.90]	0.8299 [0.81, 0.85]	0.9644	0.5948 [0.55, 0.66]	−0.369
0.50	0.7540 [0.71, 0.79]	0.7481 [0.71, 0.77]	0.9098	0.5236 [0.46, 0.59]	−0.38
0.75	0.7517 [0.72, 0.78]	0.7200 [0.68, 0.75]	0.8439	0.3545 [0.31, 0.41]	−0.48

Brackets show 95% percentile intervals across the 50 splits. Procrustes residuals: orig vs. residualized $= 0.7659 \to 0.7977$ (layer 0.25), $0.9054 \to 0.9266$ (layer 0.5), $0.9554 \to 0.9672$ (layer 0.75); $\Delta_{\mathrm{Proc}} = +0.03, +0.02, +0.01$ respectively, smaller still than the in-sample Study B2 deltas.

What Study B3 establishes.

1. **Study B2s in-sample numbers were inflated by overfitting.** The held-out variance removed is 0.72–0.88, not 0.94–0.97. The cross-validated CKA collapse is −0.37 to −0.49, not −0.57 to −0.63. The methodological finding is real but materially smaller in magnitude than the in-sample numbers suggested.

2. **The cross-validated negative finding still holds.** Lexical and template features still explain a substantial majority of the LLM activation variance (held-out 0.72–0.88 across both models and all layers), and CKA still collapses substantially after lexical residualization (from 0.84–0.96 to 0.35–0.59 on held-out test rows). The qualitative conclusion that lexical structure dominates the cross-model representational similarity survives the cross-validation.

3. **The residual cross-model signal that survives lexical control is larger than Study B2 suggested.** Cross-validated residualized CKA at layer 0.25 is 0.59 (95% CI $[0.55, 0.66]$), compared to 0.39 in the in-sample Study B2 result. There is a non-trivial cross-model representational similarity that survives proper lexical/template control. Whether this surviving similarity is operator-theoretic substrate, training-distribution residue, deeper syntactic structure, or pooling artifact cannot be discriminated from the present controls alone.

4. **Procrustes residual is the more robust metric to residualization than CKA.** The Procrustes residual change across in-sample $\rightarrow$ cross-validated is small ($+0.07 \rightarrow +0.03$ at layer 0.25); the CKA change is larger ($-0.57 \rightarrow -0.37$). Procrustes operates in a constrained k-dimensional PCA subspace and is less sensitive to the global isotropic-scaling structure that lexical features capture; CKA measures global geometric similarity and is more sensitive to it.

Methodological lesson. A lexical-residual control *must* be cross-validated. The in-sample residualization in Study B2 overstates the magnitude of the negative finding by 10–25% depending on layer and metric. For future cross-model representation comparison studies that use lexical residualization as a control, the protocol should be: (i) hold out a fraction of the prompt set, (ii) fit the lexical-to-activation regression on the training rows, (iii) compute residualized similarity metrics on the held-out rows only, (iv) average over many splits with confidence intervals reported. The Study B2 protocol should not be cited as a diagnostic standard.

8.5 Study B4 Grouped Lexical Residualization

Study B3's cross-validated residualization performed random 70/30 splits of the 600-probe set. Random splits are honest at the prompt-row level but leak template-family structure: held-out rows almost always share a template family with training rows, because each template was sampled approximately uniformly across the random split. The review of the previous version of this manuscript correctly flagged this as the next decisive control to run.

Study B4 is the grouped cross-validation. Three grouping schemes are implemented:

- **Leave-One-Template-Family-Out (LOTFO):** the 600 probes are partitioned into 28 template families (recovered post-hoc by exact substring matching against the deterministic generators in `scripts/build_phase1_probes.py`). For each template family, fit ridge on the other 27 families, residualize the held-out family's prompts, compute metrics on those residuals.

- **Leave-One-Domain-Out (LODO):** hold out one of {math, logic, polecon} entirely; train on the other two.
- **Two-vs-one split:** train on math+logic, test on polecon (the largest single grouped split available).

For each grouped split, training-time TF-IDF and template-only vocabularies are fit on the training rows only; held-out test rows are featurized using the training vocabulary and IDF weights, with out-of-vocabulary test tokens dropped. This is the proper way to extend lexical featurization to grouped held-out rows.

Result. Table 8 reports LOTFO. Table 9 reports LODO and the two-vs-one split.

Table 8: Study B4 LOTFO (28 leave-one-template-family-out splits). Variance-removed 95% percentile intervals frequently span zero and reach substantially negative values — the lexical regression fitted on 27 template families does not generalize to the held-out family. CKA is essentially unchanged by residualization: $\Delta_{\text{CKA}} = -0.006, +0.019, +0.010$ across layers.

Layer	Var. removed (A)	Var. removed (B)	CKA orig	CKA resid	CKA
0.25	0.32 [−0.44, 0.71]	0.22 [−0.50, 0.59]	0.808	0.802	−0.006
0.50	0.00 [−0.90, 0.54]	0.06 [−0.78, 0.53]	0.705	0.723	+0.019
0.75	−0.01 [−0.98, 0.59]	0.01 [−1.07, 0.58]	0.600	0.610	+0.010

Table 9: Study B4 LODO and two-vs-one split. Held-out variance removed is uniformly negative (the lexical regression makes worse-than-mean predictions on the held-out group), and CKA *increases* after residualization in every cell, by as much as +0.34. The lexical residualization fails outright when the held-out prompts share no template family with the training prompts.

Held-out	Layer	Var. rem. (A)	Var. rem. (B)	CKA orig	CKA resid	CKA	Proc
Leave-One-Domain-Out							
logic	0.25	−0.49	−0.45	0.816	0.858	+0.042	−0.006
math	0.25	−1.12	−1.20	0.904	0.948	+0.043	−0.024
polecon	0.25	−0.31	−0.61	0.774	0.815	+0.041	−0.033
logic	0.50	−0.64	−0.46	0.668	0.770	+0.103	−0.002
math	0.50	−1.93	−1.29	0.797	0.914	+0.117	+0.005
polecon	0.50	−1.41	−1.06	0.556	0.891	+0.335	−0.015
logic	0.75	−0.78	−0.44	0.480	0.621	+0.141	−0.001
math	0.75	−1.82	−1.44	0.635	0.820	+0.186	+0.001
polecon	0.75	−1.37	−0.98	0.558	0.859	+0.301	−0.011
Two-vs-one (train math+logic, test polecon)							
polecon	0.25	−0.31	−0.61	0.774	0.815	+0.042	−0.033
polecon	0.50	−1.41	−1.06	0.556	0.891	+0.335	−0.015
polecon	0.75	−1.37	−0.98	0.558	0.859	+0.301	−0.011

What Study B4 establishes. The grouped cross-validation forces a substantial reframing of the negative methodological finding from Studies B2, B3.

1. **The lexical structure that random-split residualization removed is template-specific.** The TF-IDF and template-only featurization fit on 27 template families does not generalize to the held-out template family: variance-removed point estimates hover near zero with 95% percentile intervals that extend substantially negative. Under leave-one-domain-out the regression is even worse (variance-removed strongly negative across the board: the lexical model fitted on two domains makes worse-than-mean predictions on the held-out domain).

2. **Under grouped CV, lexical residualization barely affects CKA — and often *increases* it.** LOTFO Δ_{CKA} ranges from -0.006 to $+0.019$ across layers (essentially zero). LODO and two-vs-one Δ_{CKA} are positive in every cell, ranging from $+0.04$ to $+0.34$. The CKA increase is partly a consequence of the residualization failure: when ridge regression produces worse-than-mean predictions, the residuals contain correlated noise that, when present in both models, can *increase* apparent similarity. We therefore do not read the LODO/two-vs-one $\Delta_{\text{CKA}} > 0$ as evidence of "lexical residualization strengthening alignment"; we read it as "lexical residualization fails outright in this regime."

3. **Procrustes residual is essentially unchanged under grouped CV.** The Procrustes residual changes are at most ± 0.04 in any cell of LOTFO/LODO/two-vs-one. Combined with the residualization failure, this is consistent with "the lexical regression is too template-specific to remove meaningful variance from out-of-template activations."

4. **The lexical contribution to cross-model alignment is not as large as Studies B2/B3 suggested.** The B3 cross-validated residualization removed a substantial portion of CKA (-0.37 to -0.49 on held-out random splits). Under grouped CV the same residualization removes essentially none of CKA. The honest reading is that B3's effect was partly an artifact of within-template lexical structure that random splits leak across train and test.

Methodological lesson, expanded. Random-split lexical controls in cross-model representation studies overstate the contribution of lexical structure to apparent similarity. Grouped cross-validation is the necessary strict version. But Study B4 also shows the limits of TF-IDF-based lexical featurization itself: a 455-dimensional bag-of-words featurization does not generalize across template families in a way that supports clean residualization on held-out templates. Future cross-model representation studies that use lexical residualization as a control should: (i) use domain-general lexical features (subword tokenizer counts, pre-trained sentence-encoder representations, character-n-gram TF-IDF fitted on a large corpus rather than the probe set itself), (ii) report grouped cross-validation by template family and domain explicitly, (iii) report the variance-removed statistic for the held-out group to verify the lexical regression actually generalizes before interpreting the residualization result.

What this means for the paper's central claim. The strong reading of Studies B1/B2/B3 was: most of the cross-model representational similarity at these layers is shared lexical and template encoding. Study B4 weakens this claim. The honest reading after Study B4 is: *a* *substantial portion of the cross-model similarity is template-specific encoding that random-split residualization can remove, but the structure that survives a template-grouped lexical control is much larger than the B3 number indicated.* The cross-model alignment is more robust to grouped lexical control than B3 suggested. This does not rescue the strong operator-theoretic interpretation: the surviving signal could still be training-distribution residue, deeper syntactic structure, or other shared encoding that our lexical featurization fails to capture. But the negative finding from B1/B2/B3 must be read with the B4 qualification: random-split lexical controls overstate the lexical contribution.

8.6 Study C Train/Test Procrustes Split

Study C addresses the same-data fit/eval confound by performing 50 random 70/30 splits of the Phase 1 probe set, fitting Procrustes on the training rows and evaluating the residual on the held-out test rows. PCA is fit on the combined train+test data so that the projection basis is consistent across the split. We additionally run cross-domain splits (fit on one domain, evaluate on another) at layer fraction 0.25.

Table 10: Study C train/test Procrustes split, 50 random 70/30 splits per layer fraction. The train/test gap is small at every layer, indicating that the Procrustes rotation is not overfitting the same-data fit.

Layer	Train residual (mean ± std)	Test residual (mean ± std)	Original (full set)	Gap
0.25	0.7665 ± 0.0010	0.7681 ± 0.0023	0.7668	+0.0016
0.50	0.9054 ± 0.0007	0.9064 ± 0.0017	0.9056	+0.0010
0.75	0.9555 ± 0.0003	0.9558 ± 0.0008	0.9555	+0.0004

Table 11: Study C cross-domain Procrustes generalization at layer fraction 0.25. Fit on the rows of one domain, evaluate on rows of another.

Train → Test	k	Train residual	Test residual	Gap
logic → math	8	0.7792	0.8110	+0.0318
logic → polecon	27	0.7821	0.8377	+0.0555
math → logic	8	0.7566	0.8453	+0.0887
math → polecon	8	0.7402	0.7821	+0.0419
polecon → logic	27	0.7760	0.8315	+0.0555
polecon → math	8	0.7635	0.7960	+0.0324

What Study C finds. The train/test gap on random 70/30 splits is essentially zero (0.0004 to 0.0016) at every layer fraction. The Procrustes rotation generalizes to held-out probes from the same distribution. Cross-domain generalization is also small to modest (gaps of 0.03 to 0.09): a rotation fit on one domain applies to another domain at slightly higher residual but well within the same magnitude. Combined with the fact that the original fit-and-evaluate-on-same-set residual essentially equals the held-out-test residual, this means the Procrustes alignment is a genuine property of the data, not a sample-fit artifact. Study C removes one of the methodological concerns the critique raised. It does not address the lexical-baseline concern (Study B1).

8.7 Study F Centered Kernel Alignment (CKA)

Study F's CKA piece addresses the methodological commitment in §4 that the pilot did not satisfy. We compute linear CKA [9] between the Phase 1 model activations at each pre-registered layer fraction, against a 1000-sample permutation null. Linear CKA is invariant to orthogonal transformations and isotropic scaling, providing a complementary metric to Procrustes residual.

Table 12: Study F-CKA: linear CKA between Phase 1 model activations. Full set ($S = 1000$) and per-domain ($S = 500$) permutation null samples. CKA is in $[0, 1]$; higher = more similar; permutation null mean ≈ 0.01 on the full set (and ≈ 0.05–0.10 on the smaller per-domain subsets).

Stratification	Layer	CKA	Perm. null mean	Above p_{99}?
full set ($N = 600$)	0.25	0.9636	0.0072	yes
full set	0.50	0.9109	0.0086	yes
full set	0.75	0.8427	0.0093	yes
math ($n = 200$)	0.25	0.9041	0.0318	yes
math	0.50	0.7969	0.0318	yes
math	0.75	0.6348	0.0286	yes
logic ($n = 200$)	0.25	0.8160	0.0621	yes
logic	0.50	0.6675	0.0637	yes
logic	0.75	0.4798	0.0428	yes
polecon ($n = 200$)	0.25	0.7736	0.0705	yes
polecon	0.50	0.5559	0.0427	yes
polecon	0.75	0.5575	0.0321	yes

What Study F-CKA finds. CKA is high on the full set (0.84–0.96) and substantially above the permutation null (≈ 0.01) at every layer fraction. This confirms, by an alignment-free metric, the strong representational similarity that the Procrustes residual indicated. CKA agrees with Procrustes that the LLM–LLM activation geometries are not random: they share substantive structure.

The per-domain CKA values are systematically lower than the full-set values across every domain and every layer (math 0.63–0.90, logic 0.48–0.82, polecon 0.56–0.77, vs. 0.84–0.96 for the full set). This is consistent with Study B2s finding: a substantial portion of the LLM–LLM similarity is captured by template / cross-domain prompt-identity structure that, when stratified to a single domain, diminishes. The within-domain ordering (math > logic > polecon at most layers) tracks the templating density of the domain — math probes are the most structurally repetitive (templated arithmetic) and produce the highest within-domain representational similarity, which is exactly what the lexical/template-driven reading of the alignment would predict. CKA does not by itself address the lexical-baseline concern any more than Procrustes does; it confirms the global geometric similarity is real, but the question of whether that similarity exceeds shared-language-encoding is the same question Study B answers (negatively, in the present pilot).

8.8 Joint Reading of the Initial Studies

Read together, the six initial studies produce a more nuanced picture than any single one of them gives on its own. The picture, briefly: the pilot's apparent cross-model alignment is real; a substantial component of it is template-specific lexical encoding that random-split controls can remove; under grouped (template-out) cross-validation the lexical component is much smaller than random-split residualization indicated; and the surviving cross-model similarity remains substantial without licencing the strong operator-theoretic interpretation.

The pilot result (Phases 0 and 1) and Study F-CKA agree that two architecturally different frontier transformers exhibit substantial representational similarity on matched compositional-evaluation prompts (CKA 0.84–0.96, Procrustes residual far below the permutation null). Study C confirms this is not a same-data fit artifact: train/test gaps are essentially zero. So the alignment is a genuine property of the models on the data.

Study B1 (lexical/template baselines) and Study B2 (in-sample lexical residualization) gave a strong-looking negative methodological finding: the LLM–LLM Procrustes residual is matched or exceeded by TF-IDF/template-only baselines, and 94–97% of LLM activation variance is predictable from 455 TF-IDF + template features in sample. Study B3 (cross-validated random-split residualization) moderated this finding: held-out variance removed is 72–88% (in-sample 94–97% was inflated by overfitting), and the held-out CKA collapse is -0.37 to -0.49 (in-sample -0.57 to -0.63). Even after the moderation, the pilot's apparent CKA was substantially explained by lexical structure under random-split controls.

Study B4 (grouped cross-validation: leave-one-template-family-out and leave-one-domain-out) further moderates the finding. Under LOTFO, Δ_{CKA} ranges from -0.006 to $+0.019$ (essentially zero); the lexical regression fit on 27 template families does not generalize to the 28th. Under LODO, the lexical regression makes worse-than-mean predictions on the held-out domain, and CKA *increases* after residualization (an artifact of correlated noise from regression failure). The honest reading: the lexical structure that random-split residualization removed in Study B3 was substantially template-specific. The cross-model representational similarity that survives a properly grouped lexical control is much larger than B3 indicated.

Study A independently complicates the operator-theoretic reading. PCA-derived candidate subspaces are partially preserved under the forward operator (more invariant than random subspaces in every cell), but absolute leakage is large (0.51–0.89), and the layer at which Procrustes alignment is strongest (layer 0.25) is the layer at which Phi-4's PCA invariance is *weakest*. Cross-architectural alignment and within-architecture invariance are anti-correlated across depth. The simplest operator-theoretic reading does not predict this.

The methodological finding the paper now makes. A row-permutation null is too weak a baseline for cross-model Procrustes/CKA tests on language-model activations. Lexical and template baselines (Study B1) match or exceed the LLM–LLM alignment. Random-split lexical residualization (Studies B2, B3) removes a substantial fraction of CKA, but this removal is partly template-specific: under grouped cross-validation by template family or by domain (Study B4), the same lexical residualization barely affects CKA, and TF-IDF + template-only featurization fails to generalize across template families.

The compound methodological lesson is therefore stronger than the single-study lesson would have been:

- Row-permutation nulls are insufficient. Lexical baselines (Study B1) are required.
- In-sample lexical residualization (Study B2) overstates the lexical contribution. Cross-validated residualization (Study B3) is required.
- Random-split cross-validation is insufficient. It leaks template-family structure across train and test, so the lexical regression captures within-template patterns that do not generalize. Grouped cross-validation by template family and by domain (Study B4) is required.
- Bag-of-words / template TF-IDF features are themselves insufficient. They do not generalize across template families well enough to support clean residualization on held-out templates. Future studies should use domain-general lexical features (subword tokenizer counts, large-corpus character n-grams, pre-trained sentence encoders).

The cross-model representational similarity, after the chain of controls, is materially larger than Study B2 alone indicated. Whether the surviving similarity is operator-theoretic substrate, training-distribution residue, deeper syntactic structure, or artifact of the methodology itself cannot be discriminated by these controls alone. The operator-theoretic program remains a research agenda; the pilot does not advance its strong claim; the methodological controls produce a more nuanced rather than uniformly negative picture.

What this means for the paper. The most defensible contribution of this paper, after the initial studies, is no longer "we found cross-architectural operator-subspace isomorphism." It is:

> *A row-permutation Procrustes test on cross-model frontier-transformer activations can produce extremely significant alignment results that appear, under in-sample lexical residualization, to be largely lexical; but the same alignment, under grouped cross-validation by template family or by domain, is essentially unaffected by the same lexical residualization. The lexical contribution detected by random-split controls is substantially template-specific. Future cross-model representation studies should report row-permutation z-scores, lexical baselines, in-sample lexical residualization, random-split cross-validated residualization, AND grouped cross-validation by template family and domain before claiming mechanistic, compositional, or operator-theoretic content. The chain matters: each step in the chain materially changes the interpretation of the previous step.*

This is the framing the remainder of the paper now adopts.

9 Discussion

9.1 What the Pilot Shows

The pilot shows that, on matched compositional-evaluation prompts, the PCA-projected last-token activation matrices of two architecturally different frontier transformers admit a substantially better orthogonal alignment than they do under random row permutation. This holds at three pre-registered layer fractions, in two architecturally-different model pairs (Phi-4 / Qwen3-4B in Phase 0, Phi-4 / Qwen3-Next-80B in Phase 1), and within each of three distinct compositional-evaluation domains in Phase 1 (math, logic, political-economy). The result is real. It is reproducible. The infrastructure that produces it is released for reuse and extension.

9.2 What the Pilot Does Not Show

The pilot does not show that the aligned PCA subspaces are operator-invariant in the sense of Eq. (2). The paper defines that statistic and does not measure it. PCA top-k components identify high-variance directions; high-variance is not invariance. Until the leakage statistic is computed (Study A, §7), the operator-theoretic interpretation of the alignment remains a hypothesis the pilot motivates rather than confirms.

The pilot does not show that the alignment isolates compositional structure from lexical, template, syntactic, or tokenizer-induced structure. The row-permutation null discriminates "the models encode the prompts at all" from "the models encode them identically up to an orthogonal transformation in their respective PCA subspaces." Two competent language models given the same English prompts will produce matched-row activations more similar than scrambled-row activations *regardless* of any operator-theoretic substrate, because both models encode prompt identity, lexical content, and template structure. Until lexical, template, and syntax-scrambled baselines have been run (Study B1), the pilot result is consistent with both the operator-theoretic interpretation and the much weaker shared-language-geometry interpretation.

The pilot itself did not show that the alignment is robust to held-out prompts; the original procedure fit and evaluated Procrustes on the same probe set. The first-pass Study C reported in §8.6 addresses this concern directly: train/test gap is essentially zero (≤ 0.002) for random splits and small (≤ 0.09) for cross-domain splits.

The pilot itself did not compute the promised CKA cross-validation; the first-pass Study F-CKA reported in §8.7 addresses this commitment: CKA is high (0.84–0.96 on the full set) and far above the permutation null. The alignment that Procrustes identifies is corroborated by an alignment-free metric. The methodological commitment is now satisfied for Phase 1.

The pilot, even augmented by Studies C and F-CKA, does *not* show that the alignment exceeds simple lexical and template baselines. The first-pass Study B reported in §8.2 shows the opposite: the LLM-LLM Procrustes residual is essentially indistinguishable from one of the LLMs aligned against a TF-IDF representation of the same prompts, and is worse than a TF-IDF/template-only alignment. The follow-up Study B2 (§8.3) reports the result of explicitly residualizing lexical/template structure out of the LLM activations and re-running the alignment. We treat the joint B1/B2 finding as the binding evidentiary constraint of the present paper.

The pilot does not show that the alignment survives the shared training-distribution confound. Both models in each pair are trained on heavily overlapping post-2024 web corpora. The aligned subspace could plausibly be a residue of shared corpus statistics. Until a training-distribution-disjoint control is run (Study D), the substrate-independence reading is one of two possible readings of the result, the other being shared-corpus residue.

The pilot does not show that the aligned subspace is causally responsible for compositional behavior. The alignment is correspondence, not causation. The Procrustes transformation R^* predicts which direction in V_B corresponds to a given direction in V_A, but that prediction has not been tested by intervention (Study E).

9.3 What Would Make This an Operator-Theoretic Confirmation

A subsequent paper that completes Studies A through G of §7 and reports them jointly with the pilot result would, if the results are positive, support the strong operator-theoretic claim. We do not estimate the probability that the strong claim survives the controls; the present pilot is consistent with both the strong claim and several weaker alternatives, and the controls are designed precisely to discriminate.

A version of the controls that finds:

- Low invariance leakage on the recovered subspaces (Study A positive).
- Alignment quality substantially exceeding lexical and template baselines (Study B positive).
- Robust train/test alignment with cross-domain transfer (Study C positive).
- Alignment preserved under substantial training-distribution disjointness (Study D positive).
- Cross-model intervention predictions confirmed (Study E positive).

would constitute the publication claim that this paper's title was originally written to assert. The honest current status is that the claim is hypothesized, the test infrastructure exists, the pilot result is consistent with the hypothesis, and the discriminating controls remain to be run.

9.4 Phase 2: Training-Distribution Control (the Binding Outstanding Confound)

After the joint B1/B2 lexical control reported in §8, the binding outstanding confound is shared training-distribution overlap (Study D in §7). The pilot is run on two model pairs whose training distributions overlap substantially; until that overlap is broken in a controlled way, even an alignment that survives lexical residualization is consistent with shared-corpus residue rather than substrate-independent computation. The control requires a model whose training distribution is substantially disjoint from the Phase 1 pair's. Candidate controls:

- A model trained on a non-web-pretrained corpus (e.g., a scientific-corpus model from the Galactica [60] or PMC-LLaMA family).
- A multilingual model whose training distribution is dominated by a non-English language (e.g., a French-pretrained decoder-only model).
- A small-scale model trained from scratch on a curated curriculum with documented disjointness from the pilot pair's training data.

If alignment survives the training-distribution swap, the substrate-independence reading is meaningfully supported. If alignment quality tracks training-distribution overlap (high alignment for similar training, low alignment for disjoint training), the pilot signal is reattributed to corpus-shared statistics and the operator-theoretic interpretation is correspondingly weakened.

9.5 Phase 3: Causal Intervention

A positive COSI alignment, even after Phase 1 and Phase 2, is correspondence evidence. The full operator-theoretic claim—that the aligned subspace is the locus where the compositional reasoning is *performed*, not merely *represented*—requires causal intervention. The proposed test:

1. Identify a direction $v_A \in V_A$ that, when ablated, degrades a specific compositional-reasoning behavior in M_A.
2. Use the Procrustes transformation R^* to map v_A to its correspondent $v_B = R^{*-1} v_A \in V_B$.
3. Predict the ablation effect of v_B in M_B. Run the intervention.
4. Compare predicted to observed effect.

A successful Phase 3 result—ablation effects in M_B that match those predicted from interventions in M_A via the Procrustes map—would constitute the strongest possible empirical evidence for the operator-theoretic framing of Proposition 1. We treat Phase 3 as outside the scope of the present paper and as the natural follow-up program.

9.6 Implications, Conditional on Joint Completion of Studies A–G

We separate the conditional implications from the present pilot evidence by stating them only as conditionals. *If* the program of §7 is jointly completed and returns positive results, then several consequences follow. The SAE feature dictionary [1, 2, 3, 4, 35] becomes interpretable as the empirical recovery of bases for approximately invariant subspaces of trained transformer forward operators. Cross-model transfer of mechanistic understanding becomes geometrically tractable via the recovered Procrustes transformation. The Platonic Representation Hypothesis [10] acquires a mathematical structure that explains why convergence should occur and what its limits are. The substrate-independence intuition descending from Marr [18] acquires sharp empirical operationalization on language-model substrates.

None of these consequences is established by the pilot. We list them to motivate the studies the pilot does not perform.

9.7 Authorship Note

This paper is the work of one human author and one language model in extended dialogue. The model has no continuity across sessions; the human has the operator-theoretic intuition the paper articulates and holds responsibility for the editorial and methodological judgments. Neither party, alone, would have produced this paper. The question of how authorship should be attributed in such a configuration is itself a question the field will need to answer. The author block makes our choice explicit; we recognize it is contestable.

Closing Note

The pilot is a representational-alignment result against a row-permutation null, followed by a sequence of controls that materially revise one another. Study B1 shows that lexical and template baselines can match or exceed the raw LLM–LLM Procrustes alignment. Study B2 shows that in-sample lexical residualization removes most apparent similarity, but Study B3 shows this estimate is inflated by overfitting, and Study B4 shows that random-split residualization leaks template-family structure: under grouped template/domain holdouts, TF-IDF/template residualization fails to generalize and leaves much of the cross-model similarity intact.

The paper's defensible contribution is methodological. Cross-model activation-alignment claims require a hierarchy of controls (row-permutation null, lexical baseline, in-sample residualization, random-split cross-validation, grouped cross-validation by template family and by domain, and ideally domain-general lexical featurization that itself generalizes across templates) before any mechanistic, compositional, or operator-theoretic interpretation is licensed. The COSI program remains a motivated research agenda. The present pilot neither confirms nor refutes its strong claim; it shows that the standard methodology is too underdetermined to do either.

Three follow-ups would be decisive. Stronger lexical featurization (Study B5: subword tokenizer counts, large-corpus character n-grams, pre-trained sentence encoders — features that should survive grouped template holdout). Training-distribution-disjoint controls (Study D). Cross-model causal intervention (Study E). None alone is sufficient. The discipline of running them in sequence and watching each control change the previous step's interpretation is itself part of the contribution.

10 Code and Data Availability

All code, configuration, probe data, and experimental artifacts described in this paper are publicly released at:

https://github.com/ascendanti/cosi-pilot

A Zenodo deposit assigning a DOI to the activation-matrix dataset ($\sim$ 57 MB across Phase 0 and Phase 1) and to the pre-registration packet is to be created from the same repository before peer-reviewed journal submission. The repository is organized at the following relative paths:

- `src/sovereign/research/cosi/extract.py` — residual-stream activation extraction harness, with explicit per-architecture forward-pass replication for `phi3`, `qwen3`, and `qwen3_next` dispatches.
- `src/sovereign/research/cosi/align.py` — Procrustes alignment, permutation null, random-rotation null, and shared-subspace projection.
- `scripts/cosi_smoke.py` and `scripts/cosi_smoke_80b.py` — smoke tests validating the three architecture dispatches.
- `scripts/cosi_phase0.py` — the end-to-end Phase 0 runner. The exact run reported here is archived at `runs/cosi/phase0_20260506T230920Z/`, including per-prompt activation matrices for both models, the per-layer Procrustes residuals, and the full permutation-null sample distributions.
- `data/cosi/probe_set_frontier_v1.jsonl` — the 96-prompt probe set used in Phase 0.
- `docs/research_notes/2026-05-06_cosi_design.md` — the pre-registration document declared before Phase 0 was run.
- `docs/research_notes/2026-05-06_cosi_phase0_result.md` — the post-run result note with pre-registration compliance check.

The repository is structured to make the Phase 0 result independently re-runnable. The activation matrices are deterministic conditional on model checkpoints (which are immutable HuggingFace MLX-community artifacts) and prompts (which are committed verbatim). The permutation-null sample distributions are reproducible under the seeds recorded in the result manifest. Phase 1 specifications, including the extended probe set construction protocol, are recorded in the same directory structure as they are produced.

Model checkpoints used: `mlx-community/Phi-4-reasoning-plus-8bit` and `mlx-community/Qwen3-4B-8b` for Phase 0; `mlx-community/Qwen3-Next-80B-A3B-Thinking-5bit` for Phase 1. Both architectures route through the published `mlx_lm` model modules without modification; the COSI extractor parallels the inner forward pass rather than monkey-patching it.

The reproducibility infrastructure is described in detail in the companion methodology document at `docs/REPRODUCIBILITY_STANDARD.md`, which specifies the run-manifest, environment-fingerprint, and replication-script standards that all Sovereign experimental runs adhere to.

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open-source mechanistic interpretability community whose work, even when not directly cited, shaped the framing of the hypothesis.

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